

DIABETIC RETINOPATHY SCREENING REP

SIH 2026 | Problem No. 26038 | MathWorks AI Pipeline

Image: sample_fundus

Date: 09-Sep-2026 07:51

Quality: BORDERLINE (Score 2/3)

1. AI DIAGNOSIS

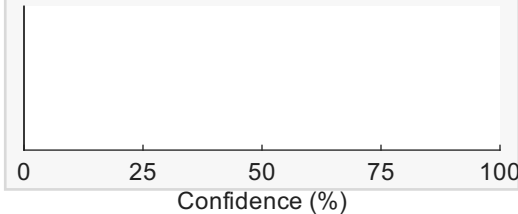
DIAGNOSIS

Moderate NPDR

Severity Level: 2 / 4 | Confidence: 92.8%

URGENT REFERRAL | Ophthalmologist tele-consult within 7 days

CONFIDENCE



2. IMAGE QUALITY METRICS

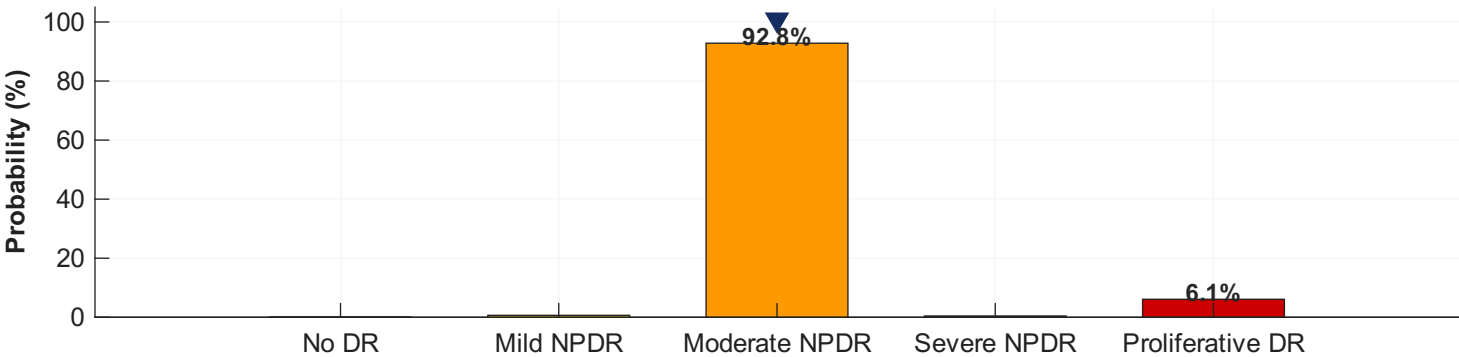
Sharpness (Laplacian Var)	FAIL 19.3
Brightness (Mean)	PASS 51.3
Contrast (Std Dev)	PASS 31.3

3. DETECTED LESION EVIDENCE

Microaneurysms (MA) Earliest sign of DR — small red-dot vascular leaks	24 found
Hard Exudates (EX) Bright yellow lipid deposits — breakdown of blood barrier	1 found
Hemorrhages (HE) Dark blotches — deeper vascular wall rupture	0 found

Optic Disc Centre: [X:371, Y:239] | Fovea Centre: [X:454, Y:244]

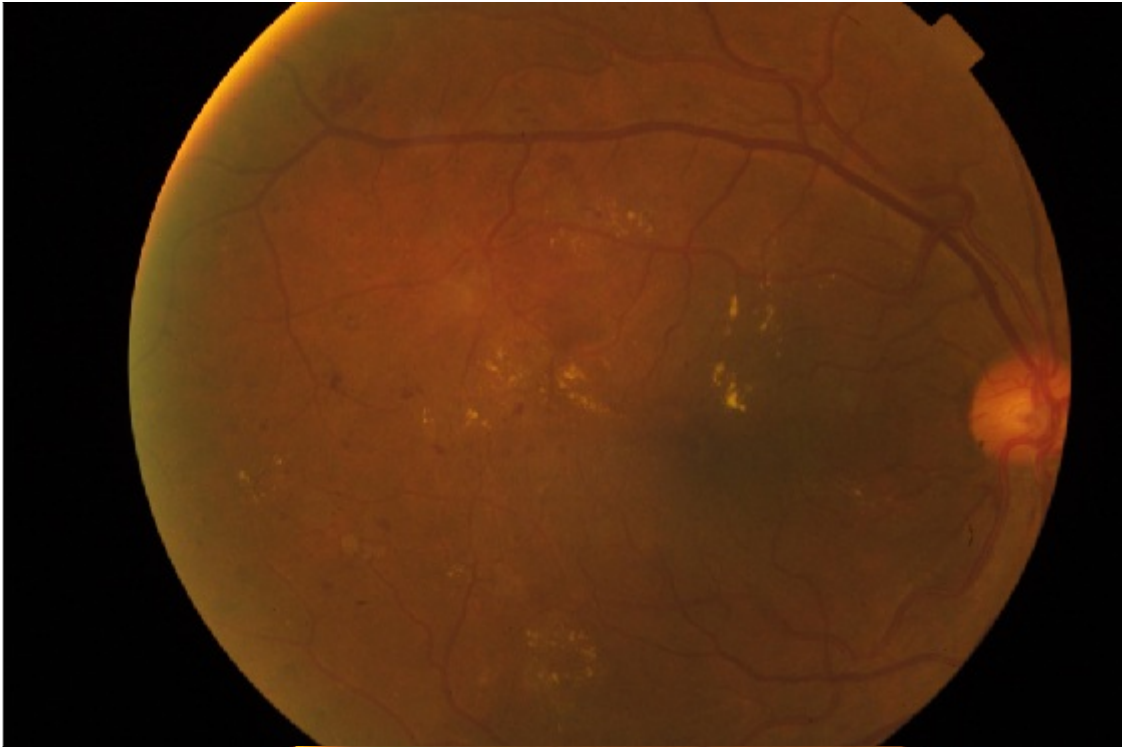
4. CLASS PROBABILITY BREAKDOWN



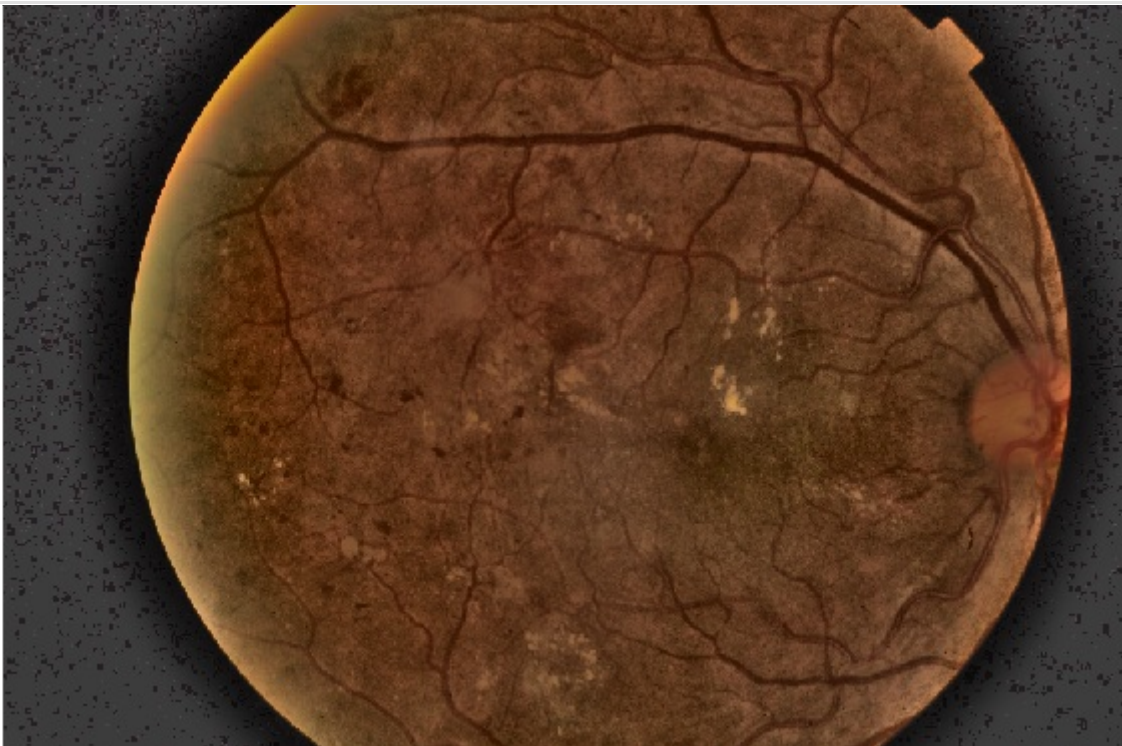
FUNDUS IMAGE ANALYSIS

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A. Original Fundus Image



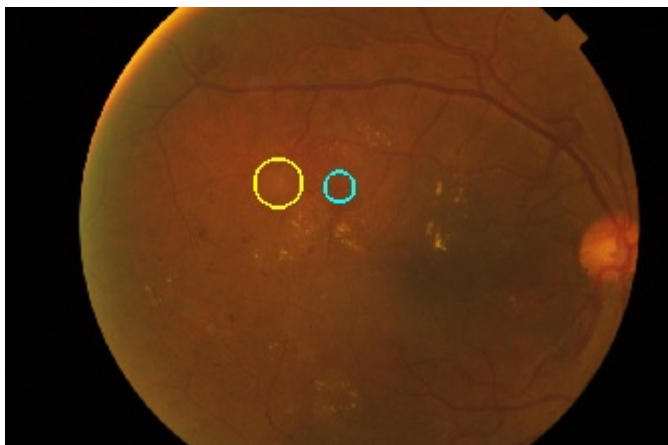
B. CLAHE Enhanced & Illumination Normalised | Quality: BORDERLINE



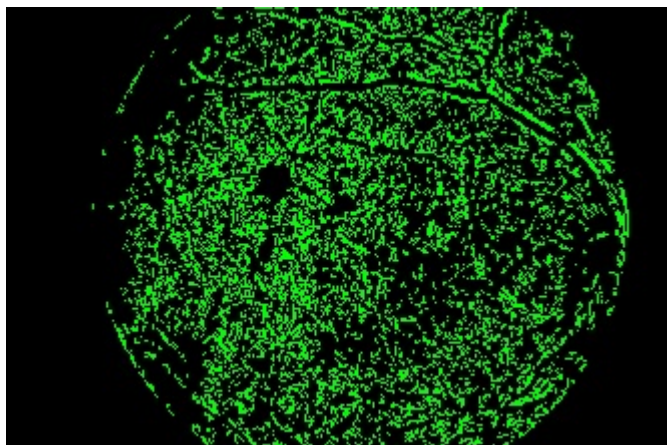
RETINAL SEGMENTATION & EXPLAINABLE AI

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A. Anatomical Landmarks [OD: Yellow | Fovea: Cyan]



B. Vessel Network Segmentation [Top-Hat Transform]



C. Lesion Map [MA=24 Red | EX=1 Yellow | HE=0 Magenta]



D. Grad-CAM Attention Map [Moderate NPDR, 92.8% Confidence]

